



STUDENT

Adithya Krishnan
R

ASSIGNMENT

ELG2 - 04 Logarithms | Code:
ELG2 | Type: AA

TOPIC

Logarithms

EVALUATION DATE

20 May 2026

97.3%

14.60 / 15 marks

Trailblazer

Below AA promotion threshold

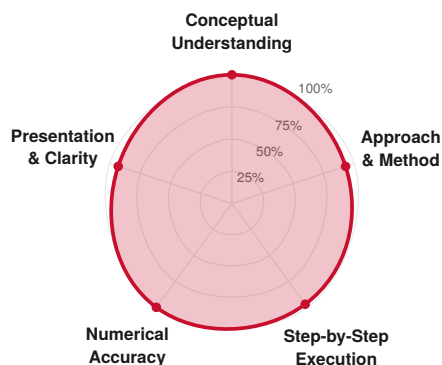
STATUS: Stays at AA tier.

Score 97.3% — 60% required to advance.

SUMMARY

Adithya scored at the top end of this assignment - effectively full marks across 15 questions, including the JEE-Advanced-level Q4, Q8, Q9, and Q13. The work shows mature habits: explicit domain constraints, rigorous case rejection, and clean identity manipulation. The only flags are a sign-line transcription slip on Q3 (which didn't affect the answer) and an over-engineered calculus route on Q15(D). The next push is method-efficiency: solve optimisation logs by algebraic factoring, not derivatives.

Rubric Breakdown



Dimension	Wt	Score	Status
Conceptual Understanding	40%	100.0%	Mastery
Approach & Method	20%	93.0%	Strong
Step-by-Step Execution	20%	97.0%	Strong
Numerical Accuracy	10%	100.0%	Mastery
Presentation & Clarity	10%	93.0%	Strong

Per-question score formula: $Q = 0.40 \times \text{CU} + 0.20 \times \text{AM} + 0.20 \times \text{SS} + 0.10 \times \text{NA} + 0.10 \times \text{PR}$, where each dimension is scored 0, 0.5, or 1. Maximum 1.00 mark per question; aggregate over 15 questions.



Concept Dependency Map

Which underlying concepts each question tests, colored by your performance. Green = full, Amber = partial, Red = failed, Grey = not tested.

Concept

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Change of base / $a^{\log_a x} = x$	Green	Green	Green	Green	Green	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Green	Grey	Grey
Domain extraction ($R_1, R_2, \dots$)	Grey	Amber	Amber	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green
Sign-line analysis for rational inequalities	Grey	Amber	Amber	Green	Green	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey
Base < 1 inequality reversal	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Amber	Grey	Grey	Grey	Grey	Green	Grey
Substitution / parameterisation	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green
Case-splitting and rejection	Grey	Amber	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green
Monotonicity / structural cancellation	Grey	Grey	Grey	Grey	Grey	Green	Green	Green	Green	Green	Green	Green	Green	Green	Amber

SWOT Matrix

STRENGTHS

- **Systematic domain extraction.** Across Q5, Q7, Q10, Q11, Q12, Q15 you consistently labelled constraints as $R_1, R_2, \dots$ before manipulating - the JEE Advanced gold standard.
- **Identity fluency under pressure.** Q13's use of $x^{\log_{16} 9} = 3^{(1/2)\log_2 x}$ and Q1's $4^{\log_2 3} = 9$ show you internalise log-exponent symmetries.
- **Rigorous case-analysis.** Q4 (rejecting $k^2 = 1$), Q7 (rejecting $x = -1, -3$), Q8 ($k > 0$ vs $k < 0$ branches) all show the discipline of pursuing then rejecting extraneous roots.
- **Structural insight on hard problems.** Q9 - spotting the vanishing factor in the product - is JEE-Advanced-tier reasoning.

WEAKNESSES

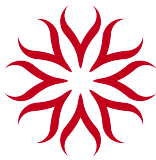
- **Transcription errors in sign lines.** Q3 wrote critical point -1 instead of $+1$ on the sign chart; happened to not affect the final answer this time.
- **Method-selection efficiency on optimisation.** Q15(D) was solved by full calculus when a 3-line algebraic factoring $-(2^t - 2)(3^t - 3) + 6$ would suffice.
- **Ambiguous final boxing.** Q2 circled both (A) and (B) due to an extra AM-GM tangent - signals uncertainty even when the work is correct.

OPPORTUNITIES

- **Ready for log-mixed transcendental problems.** Q9 and Q11 show comfort with $\sin, \tan, e^x$ embedded in log expressions - move to JEE Advanced 2020+ archive.
- **Build a 'spot the factoring' reflex.** Q13 you got it ($t^2 - 12t + 27$); Q15(D) you missed it. A focused 10-problem set on disguised factorisations will close this gap.

THREATS

- **Time pressure on Match-the-column.** Q15(D)'s calculus route, while correct, consumes minutes a JEE Advanced clock won't grant; in a real paper, the slower route may force you to skip another question.
- **Small slips on otherwise-correct problems.** Q3's -1 vs $+1$ slip and Q10's loose explanation of $\log_{0.2}$ sign-reversal are the kind of micro-errors that cost partial marks in subjective grading.



Areas of Improvement

PRIORITY

01

Critical

Factoring-pattern drill for log optimisation problems

ACTION Solve 10 problems of the form 'maximise/minimise $a \cdot b^{\log_c x} + d \cdot e^{\log_c x} - f(x)$ ' where the trick is to substitute $x = c^t$ and spot a hidden product $(p^t - p_0)(q^t - q_0)$. Avoid calculus.

MEASURE Solve each in under 4 minutes without derivative.

BY WHEN Within 1 week.

PRIORITY

02

Important

Sign-line verification protocol

ACTION Before drawing any sign line, re-read each factor and write its zero in a separate row. Then transcribe to the number line. Add a 30-second audit step: check that each labelled zero matches a factor.

MEASURE Zero transcription errors over the next 20 rational-inequality problems.

BY WHEN Within 2 weeks.

PRIORITY

03

Nice to have

Decisive answer-boxing discipline

ACTION When the question is single-answer (objective MCQ or single-value integer), commit to one box. If you discover an alternative observation mid-solution, write it as a 'Note' below the box, never in place of the box.

MEASURE Every objective answer boxed exactly once over the next assignment.

BY WHEN Next submission.



Per-Question Evaluation

Each question is worth 1 mark. The 5-dimension rubric is applied to every question and weighted to a single mark using the formula on Page 1. Each box below shows the percentage scored on that dimension (big number) with the weighted contribution to the question total below (weight $\times$ score). Sub-scores per dimension: 1 (full), 0.5 (partial), 0 (none).

Q1.

$$\log_{[A-N/10]} |N + A + D + 6| - \log_5 2 \text{ where } N = 7^{\log_{49} 900}, A = 2^{\log_2 4} + 3^{\log_2 4} + 4^{\log_2 2} - 4^{\log_2 3}, D = (\log_5 49)(\log_5 2)$$

1.00 / 1.00

Topic: Simplifying logarithmic expressions with change of base

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
2 (Option B)	2 (Option B)

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10
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Feedback: Textbook execution. You handled the $4^{\log_2 3} = 3^{\log_2 4} = 9$ symmetry cleanly, which is exactly where most students lose marks. Maintain this habit of isolating N, A, D before combining.

Q2.

$$\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b} = k$$

0.95 / 1.00

Topic: Symmetric ratios in logarithms

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$a^{b+c} \cdot b^{c+a} \cdot c^{a+b} = 1$ (also marked B)	$a^{b+c} \cdot b^{c+a} \cdot c^{a+b} = 1$ (Option A)

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 50% Weighted Score = 0.10
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Feedback: The core derivation is perfect. However, by also writing the AM-GM consequence and circling (A)(B), you signal uncertainty about which is THE answer. In objective marking, only the strictly-derived identity (A) should be highlighted. Trust your first clean derivation.



Q3. $f(x) = \log \left\{ \frac{(x-1)(x+2)}{(x-3)(x-4)} \right\}$

0.90 / 1.00

Topic: Domain of logarithm of a rational expression

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
−3 (Option B), but domain written as $(-\infty, -2) \cup (-1, 3) \cup (4, \infty)$	−3 (Option B); correct domain $(-\infty, -2) \cup (1, 3) \cup (4, \infty)$

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 50% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10
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Feedback: Right answer, but watch the transcription: you wrote a root at -1 on your sign line when the actual factor $(x-1)$ gives root $+1$. You got lucky because the largest-negative-integer answer was unaffected, but in a numerical-answer question this slip costs the full mark. Always re-read your critical points off the original factors before drawing the sign line.

Q4. $a = \log_{245} 175$, $b = \log_{1715} 875$; find $\frac{1-ab}{a-b}$

1.00 / 1.00

Topic: Change of base with two-variable parameterisation

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
5	5

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10
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Feedback: Excellent JEE-Advanced-grade work. The case analysis ruling out $k^2 = 1$ is the kind of rigour graders reward. Note your parameter $k = \log_7 5$ is the reciprocal of the key's $t = \log_5 7$ - both work and you handled it correctly.



Q5. $\log_4(2x^2 + 5x + 27) - \log_2(2x - 1) \geq 0$

Topic: Logarithmic inequality with integer count

1.00 / 1.00Scan Quality: **Good**

YOUR ANSWER		EXPECTED		
6		6		
Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10

Feedback: Model solution. The discipline of explicitly listing $R_1 : 2x^2 + 5x + 27 > 0$ (then verifying via discriminant) and $R_2 : 2x - 1 > 0$ before manipulating is exactly the JEE Advanced expectation. Carry this R_i template into every log inequality.

Q6. $\sum_{k=1}^{18} \log_{10} f(k)$ where $f(x) = (x^2 + 3x + 2)^{\cos(\pi x)}$

Topic: Sum with $\cos(\pi k)$ alternation; telescoping**0.95** / 1.00Scan Quality: **Good**

YOUR ANSWER		EXPECTED		
1		1		
Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 50% Weighted Score = 0.10

Feedback: Right answer through a genuine telescoping argument. Clean up the middle of the working: when a long product cancels, write the even- k product and odd- k product on separate lines, then divide. This will save the reader (and you, when revising) the visual effort.

Q7. $\log_{3x+4}(4x^2 + 4x + 1) + \log_{2x+1}(6x^2 + 11x + 4) = 4$

Topic: Log equation with variable base

1.00 / 1.00Scan Quality: **Good**

YOUR ANSWER		EXPECTED		
$x = 3/4$		$x = 3/4$		
Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10

Feedback: Outstanding. You stated domain $D_f = (-1/2, 0) \cup (0, \infty)$ from R_5, R_6 on the previous page, then correctly rejected $x = -1, -3$. This is the rigorous handling JEE Advanced rewards.

**Q8.****1.00** / 1.00 $\log(kx) = 2 \log(x + 2)$; count integers $k \in [-500, 500]$ with exactly one real solution

Topic: Parametric log equation with unique-root count

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
501	501

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10
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Feedback: JEE-Advanced level reasoning - the parabola sketch with $f(-2) < 0$ to justify exactly one root in the admissible interval is exactly the right move. You correctly excluded $k = 0$ (log undefined) and didn't double-count for $k > 8$.

Q9. $f(x) = \prod_{i=1}^4 (x + \ln \sin 2\alpha_i) + \tan x$; find $\sum e^{-\alpha_i} (f(\alpha_i) + 1/f(\alpha_i))$ **1.00** / 1.00

Topic: Telescoping product with tan + cot

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
8	8

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10
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Feedback: Excellent insight into the structural cancellation. Most students miss that ONE factor in the product becomes zero at $x = \alpha_i$. Your note acknowledging $\sin 2\alpha_i > 0$ (needed for $\ln$) is the kind of domain awareness that distinguishes top-tier work.



Q10. $\sqrt{1 + \log_{0.04} x} + \sqrt{3 + \log_{0.2} x} = 1$; find $\alpha^{\log_5 3}$

Topic: Sum of square-roots of logarithms

0.90 / 1.00Scan Quality: **Good**

YOUR ANSWER		EXPECTED		
9		9		
Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 50% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10

Feedback: Correct answer with proper verification. One refinement: when both $\sqrt{u^2 - 1}$ and $1 - u$ must be non-negative AND sum to 1, the cleaner argument is $u^2 - 1 \geq 0$ AND $1 - u \geq 0$ forcing $u = 1$ uniquely - no quadratic needed. Also re-check your R_3 : $\log_{0.2} x \geq -3$ with base < 1 reverses to $x \leq 0.2^{-3} = 125$ - you wrote it correctly but the explanation was muddled.

Q11. $\log_{10}(2^{x+1} + x - 37) = x(1 - \log_{10} 5)$

Topic: Mixed exponential-linear log equation

1.00 / 1.00Scan Quality: **Good**

YOUR ANSWER		EXPECTED		
1 solution ($x = 5$)		1 solution ($x = 5$)		
Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10

Feedback: Clean. The monotonicity argument ('as $2^x + x$ is increasing, only one solution') is the rigorous justification - a lot of students would just verify $x = 5$ without proving uniqueness. Well done.

Q12. $\log(x^2/y^3) = 1$, $\log(x^2y^3) = 7$; find $\log |xy|$

Topic: Simultaneous log equations

1.00 / 1.00Scan Quality: **Good**

YOUR ANSWER		EXPECTED		
3		3		
Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10

Feedback: Spot on. Recognising that $\log x^2 = 2 \log |x|$ (not $2 \log x$) is the subtle point and you handled it - excellent.

**Q13.** $3^{\log_2 x} - 12x^{\log_{16} 9} = \log_3(1/3)^{3^3}$; find ab Topic: Log equation with $x^{\log_{16} 9}$ **1.00** / 1.00Scan Quality: **Good**

YOUR ANSWER	EXPECTED
64	64

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10
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Feedback: Beautifully handled. The $x^{\log_{16} 9} = 3^{(1/2)\log_2 x}$ identity is the make-or-break step and you executed it from first principles. This is exam-clearing rigour.

Q14. $4\log_{1/4} \left(\frac{7 + \sqrt{2 + \sqrt{2 + \dots}}}{\sqrt{18}\sqrt{18\dots}} \right)$

Topic: Nested radicals inside log

1.00 / 1.00Scan Quality: **Good**

YOUR ANSWER	EXPECTED
2	2

Conceptual Understanding 100% Weighted Score = 0.40	Approach & Method 100% Weighted Score = 0.20	Step-by-Step Execution 100% Weighted Score = 0.20	Numerical Accuracy 100% Weighted Score = 0.10	Presentation & Clarity 100% Weighted Score = 0.10
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Feedback: Clean. Rejecting $x = -1$ with the reasoning 'never negative' (since $x = \sqrt{\cdot}$) is the correct domain-aware argument.

**Q15.****0.90** / 1.00(A) integral solutions of $\log_2(\log_{1/2}(1-x)) > 0$; (B) positive integers in $\log_3(1 - \log_{1/2}(x-1)) \leq 0$; (C) roots of $2 < x < 4$

Topic: Match-the-column on log inequalities and a max problem

Scan Quality: **Good**

YOUR ANSWER

EXPECTED

A-S, B-R, C-P, D-Q

A-S, B-R, C-P, D-Q

Conceptual
Understanding**100%**

Weighted Score = 0.40

Approach
& Method**50%**

Weighted Score = 0.20

Step-by-Step
Execution**100%**

Weighted Score = 0.20

Numerical
Accuracy**100%**

Weighted Score = 0.10

Presentation
& Clarity**100%**

Weighted Score = 0.10

Feedback: Full credit. One stylistic note on (D): using $f'(x) = 0$ buys you the answer but cost you minutes. The slicker route is to write $x = 6^t = 2^t \cdot 3^t$, then the expression becomes $-(2^t - 2)(3^t - 3) + 6$ which is ≤ 6 with equality at $t = 1$. Add this 'spot the product' habit to your toolkit - it saves time on optimisation-flavoured log problems.

Closing Note

Adithya, this is a near-perfect ELG2. Across 15 questions - several of them genuine JEE Advanced level - you executed with the discipline of someone who has internalised the rules of logarithms rather than memorised them. Effort and understanding are both clearly visible.

Your score reflects four habits that matter at the top tier: you extract domain constraints before manipulating, you pursue cases and reject extraneous roots cleanly, you fluently apply $a^{\log_b c} = c^{\log_b a}$ -type symmetries, and you verify final answers. These are exactly the habits JEE Advanced graders reward in subjective sections.

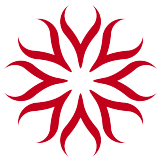
Where to go next.

- **Next assignment:** ELG3 (Quadratic Equations / Theory of Equations) - aim to finish in 90 minutes.
- **Adjacent topic:** Begin Inequalities (AM-GM, Cauchy-Schwarz) since Q2 hints you already reach for them - formalise that toolkit.
- **Speed target:** Sub-5-minute average per question on the next AA paper; flag any question crossing 7 minutes as a candidate for method-review.

AI-Assisted Evaluation Disclaimer

This report was generated with AI assistance. AI can make mistakes. If any score, feedback, or scan-quality assessment seems off, please flag it for human review. The Standard Subjective Rubric (40% Conceptual Understanding, 20% Approach & Method, 20% Step-by-Step Execution, 10% Numerical Accuracy, 10% Presentation & Clarity) was applied consistently across all questions. Final responsibility for grading decisions rests with the evaluating teacher.

Evaluated by ThinkingSouls evaluation framework · Standard Subjective Rubric (40/20/20/10/10) · All five dimensions weighted to a single mark per question.



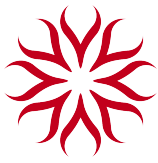
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Scan Quality Overview

Across the 15 questions: **Excellent: 0** · **Good: 15** · **Acceptable: 0** · **Poor: 0**

Scan quality affects how reliably the work can be evaluated. Each question's scan quality is logged on its individual card.